Notes: Boguth, Duchin, and Simutin (JF, 2022)

Dissecting Conglomerate Valuations

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1 Overview

Most empirical work on conglomerates needs division-level "investment opportunities" or valuation multiples. The classic approach (Lang & Stulz 1994; Berger & Ofek 1995) proxies a division's Tobin's q with the *median q of standalone firms* in the division's industry. This is convenient but imposes a strong assumption:

$$q_{k,t}^{\text{division}} \approx q_{k,t}^{\text{standalone}}.$$

Boguth, Duchin, Simutin (2022) show that this approximation fails in systematic ways and propose a method to estimate **division (industry-class) valuation multiples implied by conglomerates themselves**, without relying on standalone firms.

Main contribution (what is new). They develop a **conglomerate-only** estimator for *divisional* (industry-class) q's using cross-sectional **median (quantile) regressions** each year. This produces a time series of implied divisional q's that:

- differ materially from standalone q's across industries and time (Table II; Figure 2),
- are less sensitive to macro/industry shocks (Table IV) and especially to *credit shocks* (Table V),
- reveal a mechanism consistent with **coinsurance/internal capital markets** via covariance structure (Tables VI–VII),
- predict **internal allocation** (investment–q sensitivities; Tables VIII–IX) and **cross-firm reallocation** (diversifying M&A; Tables X–XI).

2 Notation and measurement objects

2.1 Firm value, fundamentals, and class weights

Fix a year t. Let $i = 1, \ldots, I_t$ index conglomerates and $k = 1, \ldots, K$ index "classes" (industries or industry×group cells).

- $V_{i,t}$: conglomerate i's market value object used in the multiple (the paper works with **sales multiples**).
- $W_{i,k,t}$: conglomerate i's fundamentals in class k (typically **segment sales**).
- $W_{i,\cdot,t} \equiv \sum_{k=1}^{K} W_{i,k,t}$: total fundamentals (total sales).
- $\tilde{V}_{i,t} \equiv V_{i,t}/W_{i,\cdot,t}$: conglomerate **value-to-sales** ratio (the paper calls this a q proxy).
- $\tilde{W}_{i,k,t} \equiv W_{i,k,t}/W_{i,\cdot,t}$: class k's share of conglomerate i's fundamentals.

2.2 Standalone q versus divisional q

- **Standalone** q for class k at t: the Berger–Ofek style median sales multiple among single-segment firms in class k.
- **Divisional** q for class k at t: the paper's new estimate implied by conglomerates only.

2.3 Relative divisional q (RDQ)

A central comparative statistic is the log difference:

$$\mathrm{RDQ}_{k,t} \equiv \log q_{k,t}^{\mathrm{Div}} - \log q_{k,t}^{\mathrm{SA}}. \tag{1}$$

Interpretation: $\mathrm{RDQ}_{k,t} > 0$ means conglomerate divisions in class k are valued at a premium relative to standalone firms in k; $\mathrm{RDQ}_{k,t} < 0$ indicates a "conglomerate division discount" within that class.

3 Model block 1: the new estimator for divisional q

3.1 Value additivity benchmark (paper Eq. (1))

Let $q_t^C \in \mathbb{R}^K$ be the vector of class-level valuation multiples implied for conglomerate divisions in year t. The conceptual benchmark is that conglomerate value equals the sum of class pieces:

$$V_t = W_t\, q_t^C, \tag{2}$$

which is Eq. (1) in the paper.

Economic meaning. If divisions in different classes are valued using class-specific multiples (and cross-division interactions only matter through those multiples), then a conglomerate's total value is a weighted sum of its class fundamentals.

3.2 Scaling by total fundamentals (paper Eq. (2))

Divide each conglomerate's identity by its total fundamentals to work with multiples:

$$\tilde{V}_t = \tilde{W}_t\, q_t^C, \tag{3}$$

which is Eq. (2) in the paper.

3.3 Why not OLS? Why median regression?

Sales multiples are skewed. Taking logs is not additive in (3). The paper therefore targets **medians** rather than means and estimates q_t^C by cross-sectional **quantile regression** (median regression) each year:

$$\widehat{q}_t^C \ \in \ \arg\min_{q\in\mathbb{R}^K} \sum_{i=1}^{I_t} \rho_{0.5}\left(\tilde{V}_{i,t} - \sum_{k=1}^{K} \tilde{W}_{i,k,t}\, q_k\right), \qquad \rho_{0.5}(u) = |u|. \tag{4}$$

Interpretation of identification. The regression uses variation in conglomerates' class shares $\tilde{W}_{i,k,t}$ to back out the class multiples $q_{k,t}^C$. Intuitively, long–short combinations of conglomerates can "isolate" a class exposure, so the cross section traces out the class-specific multiple.

3.4 A model-based validation step: "excess value" volatility (Figure 1)

Following Berger & Ofek (1995), define **excess value** as:

$$\mathrm{EV}_{i,t} \equiv \log(V_{i,t}) - \log(\widehat{V}_{i,t}), \qquad \widehat{V}_{i,t} \equiv \sum_k W_{i,k,t}\, \widehat{q}_{k,t}, \tag{5}$$

where $\widehat{q}_{k,t}$ is either the paper's $\widehat{q}^{\mathrm{Div}}_{k,t}$ or the traditional $\widehat{q}^{\mathrm{SA}}_{k,t}$. Figure 1 compares the *time-series standard deviation* of $\mathrm{EV}_{i,t}$ across firms for the two imputation methods. Lower dispersion indicates that imputed values track true values more stably (less "measurement noise").

4 Model block 2: industry comparisons and endogenous pairing

4.1 Industry-level divisional vs standalone valuation

Given $\widehat{q}^{\mathrm{Div}}_{k,t}$ and $q^{\mathrm{SA}}_{k,t}$, the paper studies:

- time-series correlations $\mathrm{Corr}_t(q^{\mathrm{Div}}_{k,t}, q^{\mathrm{SA}}_{k,t})$,
- time-series dispersion (std. dev. across t),
- RDQ moments (mean/quantiles over time).

4.2 Endogenous industry pairing inside conglomerates

Because conglomerates choose their industry mix non-randomly (e.g., relatedness), the paper checks robustness using within-conglomerate pairings. Conceptually:

- Panel A of Table III reports $\Pr(\text{also operates in } r \mid \text{operates in } c)$ across industries.
- Panel B recomputes RDQs after *dropping* all conglomerates that operate in a given "paired" industry, to see whether RDQs are driven by particular pairings.

5 Model block 3: determinants — economic shocks and credit shocks

5.1 Univariate sensitivity regressions (Tables IV–VI)

For each industry k, the paper runs time-series regressions of annual q changes on contemporaneous shocks. A generic form is:

$$\Delta q^{(\cdot)}_{k,t} = \alpha_k + \beta_k\, \mathrm{Shock}_t + \varepsilon_{k,t}, \tag{6}$$

where $\Delta q_{k,t}$ is normalized to unit standard deviation and "Shock" is signed so that higher values mean better conditions. They report average slopes $\mathbb{E}_k[\widehat{\beta}_k]$ and average R^2 across the 10 industries, separately for:

$$q^{\mathrm{SA}}_{k,t}, \qquad q^{\mathrm{Div}}_{k,t}, \qquad q^{\mathrm{Div}}_{k,t} - q^{\mathrm{SA}}_{k,t}.$$

5.2 Principal-component shock indices (Panels B in Tables IV–V)

Let X^{econ}_t denote a vector of standardized economic shocks (market return, dividend yield, VIX, IP growth, TFP, etc.), and let X^{credit}_t denote standardized credit shocks (TED, default spread, credit rating/spread, distance-to-default, CDS). They construct:

$$\mathrm{PC}^{\mathrm{econ}}_t = \text{1st principal component of } X^{\mathrm{econ}}_t, \qquad \mathrm{PC}^{\mathrm{credit}}_t = \text{1st principal component of } X^{\mathrm{credit}}_t.$$

5.3 Horse-race regression: economic vs credit shocks (Table VII)

They jointly regress:

$$\Delta q_{k,t}^{(\cdot)} = \alpha_k + \beta_k^{\text{econ}}\, \text{PC}_t^{\text{econ}} + \beta_k^{\text{credit}}\, \text{PC}_t^{\text{credit}} + \varepsilon_{k,t}. \tag{7}$$

Key interpretation: credit shocks absorb much of the "economic-shock" wedge between divisional and standalone sensitivities; the wedge operates mainly through a financing channel.

6 Model block 4: coinsurance and covariance structure (mechanism)

The paper argues that a key reason divisions are insulated from credit shocks is coinsurance: divisions with low covariance with the rest of the conglomerate can be supported by internal capital markets when external finance tightens.

6.1 Coinsurance potential

Operationally, they compute an industry covariance matrix using standalone firms' annual sales growth, map it to each conglomerate's industry mix, and compute each division's relative contribution to conglomerate variance.

- **High coinsurance potential** ⇒ below-median contribution to conglomerate variance.
- **Low coinsurance potential** ⇒ above-median contribution to conglomerate variance.

Then they estimate divisional q's within *industry*×*coinsurance* cells and re-run credit-shock sensitivity regressions.

6.2 Implication

If coinsurance/internal reallocation is the mechanism, then:

$$\left|\beta^{\text{credit}}_{\text{high coinsurance}}\right| < \left|\beta^{\text{credit}}_{\text{low coinsurance}}\right|,$$

i.e., high-coinsurance divisions should be much less sensitive to credit shocks (Table VI), and the difference should remain in the horse race (Table VII, Panel B).

7 Model block 5: real effects — internal capital allocation

7.1 Investment–q sensitivity (Table VIII)

Let u index an observational unit: either a standalone firm or a conglomerate division. Define the investment rate:

$$I_{u,t} \equiv \frac{\text{CAPEX}_{u,t}}{\text{Sales}_{u,t}}.$$

Let $C_u = 1$ for conglomerate divisions and $C_u = 0$ for standalone firms. A simplified version of the paper's panel regression is:

$$\begin{aligned} I_{u,t} = \alpha &+ \beta_1 q^{\text{SA}}_{k(u),t} + \beta_2 q^{\text{Div}}_{k(u),t} + \beta_3 C_u \\ &+ \beta_4\, q^{\text{SA}}_{k(u),t} \times C_u + \beta_5\, q^{\text{Div}}_{k(u),t} \times C_u + \delta\, \text{CF}_{u,t} + \text{FE} + \varepsilon_{u,t}. \end{aligned} \tag{8}$$

Interpreting marginal effects.

- For a **standalone firm** ($C_u = 0$): sensitivity to standalone q is β_1; sensitivity to divisional q is β_2.
- For a **conglomerate division** ($C_u = 1$): sensitivity to standalone q is $\beta_1 + \beta_4$; sensitivity to divisional q is $\beta_2 + \beta_5$.

The paper's key finding is $\beta_5 > 0$ and sizable, while β_2 is small: divisions respond to *divisional* q more than to standalone q.

7.2 Spinoffs as a within-entity organizational-change design (Table IX)

For entities affected by spinoffs, let $\mathrm{PRE}_{u,t}$ indicate the pre-spinoff period. They estimate:

$$\begin{aligned} I_{u,t} = \alpha &+ \beta_1 q^{\mathrm{SA}}_{k(u),t} + \beta_2 q^{\mathrm{Div}}_{k(u),t} + \beta_3 \mathrm{PRE}_{u,t} \\ &+ \beta_4\, q^{\mathrm{SA}}_{k(u),t} \times \mathrm{PRE}_{u,t} + \beta_5\, q^{\mathrm{Div}}_{k(u),t} \times \mathrm{PRE}_{u,t} + \mathrm{FE} + \varepsilon_{u,t}. \end{aligned} \tag{9}$$

Interpretation: before spinoff, the entity behaves like a division (investment tracks divisional q); after spinoff, it behaves like a standalone (investment tracks standalone q).

8 Model block 6: real effects — cross-firm reallocation via diversifying M&A

8.1 Industry-year acquisition intensity (Table X)

For industry k and year t, define:

$$A_{k,t} \equiv \#\{\text{diversifying acquisitions involving industry } k \text{ in year } t\}.$$

They estimate:

$$\log(1 + A_{k,t}) = \alpha_k + \beta_1 q^{\mathrm{SA}}_{k,t-1} + \beta_2 \mathrm{RDQ}_{k,t-1} + \varepsilon_{k,t}. \tag{10}$$

Interpretation: when RDQ is higher (divisional valuations exceed standalone), diversifying acquisitions become more frequent.

8.2 Deal-level announcement returns (Table XI)

For deal d, with acquirer a and target g, define announcement returns (in excess of market returns) over a symmetric window around the announcement:

$$\mathrm{CAR}^a_d, \qquad \mathrm{CAR}^g_d, \qquad \mathrm{CAR}^{\mathrm{combined}}_d.$$

A stylized regression is:

$$\mathrm{CAR}_d = \alpha + \beta_1 q^{\mathrm{SA}}_{\mathrm{relevant},t(d)-1} + \beta_2 \mathrm{RDQ}_{\mathrm{relevant},t(d)-1} + \Gamma' Z_d + \mathrm{YearFE} + \varepsilon_d, \tag{11}$$

where Z_d controls for size, profitability, and asset growth of both parties. **Interpretation:** positive β_2 means deals in periods/industries where conglomeration is relatively more valuable (higher RDQ) have higher announcement returns.

9 Read the figures and tables

9.1 Figure 1: excess-value volatility as a validation of the new $q^{\mathbf{Div}}$

Figure 1 plots kernel densities of firm-level $\mathrm{sd}_t(\mathrm{EV}_{i,t})$ under two imputation rules:

- **New method (divisional q):** impute using $\widehat{q}^{\mathrm{Div}}$ from conglomerates only.
- **Traditional method (standalone q):** impute using medians of single-segment firms by 4-digit SIC.

Takeaway: the distribution under the new method is shifted toward smaller standard deviations, suggesting the new divisional q's track conglomerate values more stably (less noise in "excess value").

Table I

Summary Statistics

This table reports summary statistics for the 1978 to 2018 sample of conglomerates defined on the basis of the Fama-French 10-industry classification. Sales and assets are in 2018 CPI-adjusted dollars. The second-to-last row shows the number of unique conglomerates and divisions. The last row shows the number of conglomerate-year and division-year observations.

	Conglomerates		Divisions	
	Mean	Median	Mean	Median
Sales, billion	$\boxed{5.42}$	0.99	2.43	0.32
Assets, billion	5.78	1.02	2.60	0.35
CAPEX-to-sales ratio, percent	9.07	4.23	12.21	3.78
Value-to-sales ratio (q)	1.43	0.99		
Number of segments	$\boxed{2.29}$	2.00		
Number of unique observations	2,575		$\boxed{6,427}$	
Pooled number of observations	13,033		29,809	

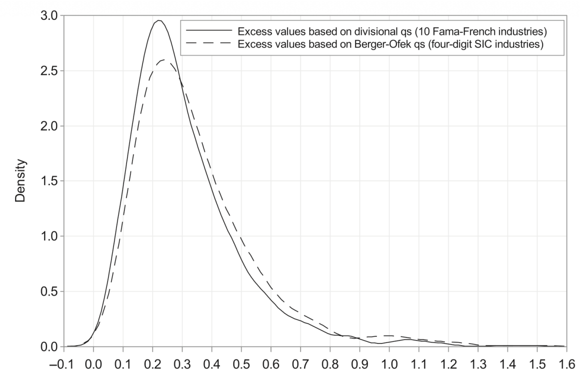

Figure 1. Kernel density of standard deviations of conglomerate excess values. The figure plots the kernel density of the time-series standard deviation of two measures of conglomerates' excess value, where excess value is the log-difference between the conglomerates' true and imputed values. Imputed values are calculated as the sales-weighted sum of either the new divisional qs obtained from quantile regressions using the 10 Fama-French industries (solid line) or the median qs of standalone firms in the respective four-digit SIC industries as in Berger and Ofek (1995) (dashed line). For each firm, the standard deviation of the timeseries of its excess values is computed, requiring at least five consecutive observations.

9.2 Table I: summary statistics

Purpose: describe the conglomerate sample and confirm it is economically comparable to the classic conglomerate literature.

What We learn:

- Conglomerates are large (mean sales $\approx$ \$5.42B) and operate in about 2.29 segments on average.
- The paper works with **sales multiples**: mean value-to-sales $q \approx 1.43$ (median 0.99).
- There are 2,575 unique conglomerates and 6,427 divisions (1978–2018).

9.3 Table II: divisional vs standalone q by industry (core descriptive evidence)

Purpose: establish that division-level valuation multiples implied by conglomerates differ from those of standalone firms *systematically*.

Main takeaways we would memorize:

- Time-series correlations between q^{Div} and q^{SA} vary widely (e.g., low in health, high in manufacturing).
- RDQs are significantly different from zero in *almost all* sectors (manufacturing is the notable near-zero case).

- Large negative RDQs appear in energy, high-tech, and health; premia appear in nondurables and telecom.

9.4 Figure 2: time series of RDQ by industry

Purpose: show RDQ is not constant over time; the "relative value of conglomeration" is state-dependent and industry-specific.

Interpretation: RDQ moves with time-varying financial/economic conditions. This motivates the later shock and financing analyses.

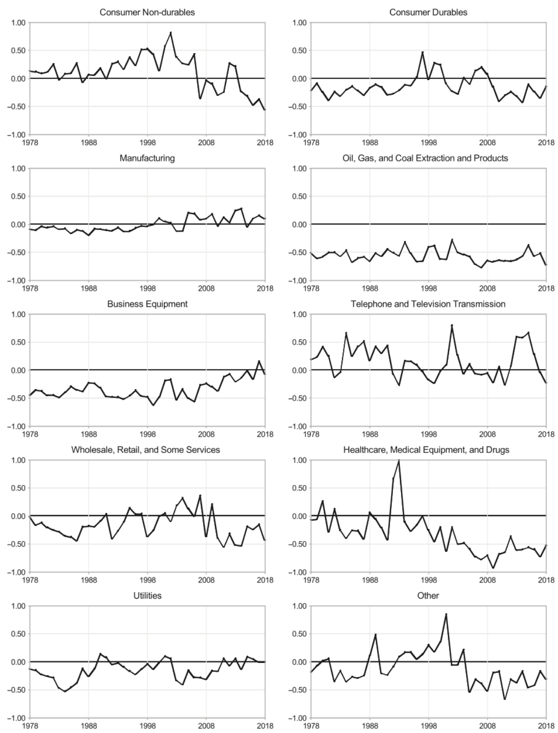

Figure 2. The timeseries of relative divisional q**s.** This figure plots relative divisional qs for divisions in the 10 Fama-French industries.

Table II

Industry Analysis of Standalone and Divisional q**s**

This table reports qs of conglomerate divisions and standalone firms, as well as their log difference (relative divisional qs, RDQs) for each of the 10 Fama-French industries. The table also reports bootstrapped t-statistics and time-series moments.

	NDur	Dur	Manu	Enrg	HiTec	Tlcm	Shop	Hlth	Util	Oth
Divisional q	0.92	0.64	0.90	1.37	1.19	3.21	0.43	1.45	1.97	1.11
Std. deviation	0.34	0.21	0.33	0.44	0.68	0.85	0.16	0.60	0.79	0.42
Standalone q	0.87	0.74	0.88	3.15	1.72	2.88	0.51	2.49	2.24	1.30
Std. deviation	0.39	0.18	0.24	0.85	0.64	0.95	0.13	1.30	0.64	0.49
Correlation	0.56	0.67	0.96	0.63	0.83	0.68	0.57	0.40	0.91	0.58
RDQ	0.12	−0.13	0.00	−0.56	−0.33	0.16	−0.16	−0.31	−0.14	−0.10
t-statistic	[3.07]	[−2.65]	[0.04]	[−37.6]	[−17.7]	[5.68]	[−3.47]	[−14.0]	[−6.02]	[−4.66]
Std. deviation	0.30	0.19	0.12	0.11	0.17	0.28	0.23	0.38	0.17	0.30
RDQ, percentile 25	−0.03	−0.26	−0.09	−0.64	−0.46	−0.06	−0.35	−0.59	−0.26	−0.31
RDQ, median	0.14	−0.16	−0.03	−0.56	−0.36	0.11	−0.18	−0.36	−0.13	−0.17
RDQ, percentile 75	0.28	−0.08	0.11	−0.50	−0.23	0.31	0.00	−0.15	−0.01	0.09
No. conglomerates	64	53	169	51	84	26	98	29	35	117

9.5 Table III: industry pairing inside conglomerates and robustness

Panel A: shows conglomerates choose nonrandom industry mixes (pairing probabilities are far from uniform).

Panel B: recomputes RDQs after omitting conglomerates that operate in specific paired industries. **Takeaway:** despite endogenous pairing, RDQ estimates are mostly stable; only about 10% of the omitted-pair cases differ significantly from baseline RDQs.

9.6 Table IV: divisional q is less sensitive to broad economic shocks

Design: industry-level univariate regressions of normalized Δq on each economic shock.

Key message:

$$|\beta^{\mathrm{SA}}| > |\beta^{\mathrm{Div}}| \quad \Rightarrow \quad (q^{\mathrm{Div}} - q^{\mathrm{SA}}) \text{ falls in booms and rises in busts (relative insulation).}$$

In practical terms, divisions appear more "buffered" against macro/industry conditions.

9.7 Table V: divisional q is less sensitive to credit shocks (stronger wedge)

Design: same structure as Table IV, but the shocks are credit/financing conditions (TED, default spread, credit rating/spread, distance to default, CDS).

Takeaway: the divisional–standalone wedge in sensitivity is large and robust. This points to a financing-frictions channel.

Table III

Industry Analysis of Divisional qs: The Role of Industry Pairings

This table shows the conditional frequency distribution of industry pairings in conglomerates (Panel A) and relative divisional qs (RDQs) estimated when industry pairings are omitted (B). For conglomerates operating in the industry indicated by the column, Panel A reports the proportion that also have divisions in the industry indicated by the row. Panel B shows average RDQs obtained by omitting from the sample all conglomerates that operate in the industry indicated by the row. *, **, and *** indicate RDQs that are significantly different from those estimated in the full sample (Table II) at the 10%, 5%, and 1% level, respectively. The bottom four rows report the average, standard deviation, minimum, and maximum of these RDQs.

	NDur	Dur	Manu	Enrg	HiTec	Tlcm	Shop	Hlth	Util	Oth
Panel A. Conditional Distribution of Industry Pairings										
NDur	1.00	0.13	0.16	0.05	0.10	0.35	0.23	0.13	0.03	0.12
Dur	0.11	1.00	0.22	0.06	0.17	0.06	0.06	0.10	0.05	0.08
Manu	0.42	0.69	1.00	0.39	0.55	0.12	0.35	0.48	0.16	0.38
Enrg	0.04	0.06	0.12	1.00	0.04	0.07	0.13	0.06	0.51	0.17
HiTec	0.13	0.27	0.27	0.07	1.00	0.26	0.13	0.28	0.05	0.15
Tlcm	0.14	0.03	0.02	0.03	0.08	1.00	0.05	0.03	0.08	0.10
Shop	0.35	0.11	0.20	0.25	0.15	0.18	1.00	0.20	0.25	0.26
Hlth	0.06	0.05	0.08	0.04	0.10	0.04	0.06	1.00	0.03	0.04
Util	0.02	0.04	0.03	0.35	0.02	0.11	0.09	0.03	1.00	0.13
Oth	0.22	0.17	0.27	0.39	0.21	0.44	0.31	0.17	0.44	1.00
Panel B. Relative Divisional qs If Industry Pairings Are Omitted										
NDur		−0.11	0.00	−0.55	−0.33	0.09	−0.10	−0.31	−0.13	−0.10
Dur	0.12		−0.04	−0.55	−0.33	0.14	−0.16	−0.30	−0.14	−0.06
Manu	0.14	−0.24		−0.50**	−0.30	0.16	−0.15	−0.40*	−0.15	−0.01**
Enrg	0.13	−0.12	−0.02		−0.32	0.15	−0.15	−0.33	−0.05***	−0.14
HiTec	0.08	−0.09	0.00	−0.57		0.26**	−0.15	−0.28	−0.13	−0.04*
Tlcm	0.02	−0.13	0.01	−0.56	−0.32		−0.13	−0.30	−0.16	−0.13
Shop	0.15	−0.16	0.06*	−0.55	−0.32	0.22		−0.26	−0.05***	−0.13
Hlth	0.09	−0.11	−0.01	−0.55	−0.36	0.17	−0.16		−0.15	−0.08
Util	0.12	−0.13	0.02	−0.57	−0.32	0.15	−0.19	−0.31		−0.09
Oth	0.11	−0.14	0.01	−0.56	−0.34	0.00***	−0.13	−0.28	−0.19	
Average	0.11	−0.14	0.00	−0.55	−0.33	0.15	−0.15	−0.31	−0.13	−0.09
St. Dev.	0.04	0.04	0.03	0.02	0.02	0.07	0.02	0.04	0.05	0.04
Min	0.02	−0.24	−0.04	−0.57	−0.36	0.00	−0.19	−0.40	−0.19	−0.14
Max	0.15	−0.09	0.06	−0.50	−0.30	0.26	−0.10	−0.26	−0.05	−0.01

Table IV

Economic Determinants of Standalone and Divisional qs

This table investigates the sensitivity of standalone and divisional qs to economic conditions. It reports average slope coefficients and R^2s from industry-level univariate regressions. The dependent variable is the annual change in q, normalized to have unit standard deviation, and the independent variable is a contemporaneous economic shock. In Panel B, the independent variables are two versions of the first principal component of the variables in Panel A based on either the variables available for the entire sample or the average first principal component that uses all combinations of available variables to maximize the estimation period. t-Statistics are reported in brackets. Data sources and construction details are provided in Appendix B. The sample period is 1979 to 2018, except for ΔVIX (1987 to 2018).

	Standalone qs		Divisional qs		Div. – SA q	
	Slope	R^2	Slope	R^2	Slope	R^2
Panel A. Economic Variables						
Market return	2.98	0.27	1.59	0.11	−1.39	0.06
	[13.34]		[4.07]		[−4.96]	
Industry return	2.44	0.39	1.03	0.12	−1.41	0.13
	[12.14]		[2.62]		[−6.61]	
−ΔDividend yield	0.66	0.08	0.28	0.02	−0.37	0.03
	[6.39]		[3.38]		[−3.74]	
−ΔIndustry dividend yield	0.42	0.04	0.21	0.01	−0.21	0.02
	[3.04]		[2.44]		[−2.55]	
−ΔVIX	0.09	0.20	0.05	0.09	−0.04	0.03
	[10.78]		[4.10]		[−5.16]	
ΔIndustrial production	6.24	0.07	3.23	0.04	−3.01	0.03
	[8.77]		[2.91]		[−2.92]	
ΔTFP	0.21	0.11	0.09	0.03	−0.12	0.04
	[7.27]		[2.92]		[−7.30]	
ΔOutput	0.08	0.05	0.04	0.02	−0.04	0.02
	[6.54]		[2.68]		[−2.68]	
ΔExpansion indicator	1.10	0.14	0.61	0.05	−0.49	0.04
	[9.30]		[5.87]		[−3.59]	
Panel B. First Principal Component of Economic Variables						
Full-sample variables	0.21	0.19	0.11	0.06	−0.11	0.05
	[11.93]		[4.50]		[−5.82]	
All variables, PC average	0.22	0.22	0.11	0.07	−0.11	0.06
	[12.48]		[4.44]		[−6.16]	

Table V

Financing Conditions: Standalone and Divisional qs

This table investigates the sensitivity of standalone and divisional qs to financing conditions. It reports average slope coefficients and R^2s from industry-level univariate regressions. The dependent variable is the annual change in q, normalized to have unit standard deviation, and the independent variable is a contemporaneous credit shock. In Panel B, the independent variables are two versions of the first principal component of the variables in Panel A based on either the variables available for the entire sample or the average first principal component that uses all combinations of available variables to maximize the estimation period. t-Statistics are reported in brackets. Data sources and construction details are provided in Appendix B. The sample period is 1979 to 2018, except for ΔTED (1987 to 2018), ΔCredit rating (1986 to 2017), ΔCredit spread (2005 to 2016), and ΔCDS (1991 to 2018).

	Standalone qs		Divisional qs		Div. – SA q	
	Slope	R^2	Slope	R^2	Slope	R^2
Panel A. Credit Shock Variables						
−ΔTED spread	1.02	0.11	0.67	0.04	−0.35	0.02
	[4.92]		[4.77]		[−2.42]	
−ΔDefault spread	0.53	0.10	0.23	0.05	−0.30	0.05
	[7.12]		[2.17]		[−2.89]	
−ΔCredit rating	0.51	0.04	0.23	0.03	−0.27	0.01
	[3.06]		[1.56]		[−2.44]	
−ΔCredit spread	0.43	0.44	0.14	0.19	−0.29	0.17
	[6.14]		[1.11]		[−2.09]	
ΔDistance to default	0.37	0.37	0.18	0.14	−0.19	0.11
	[9.47]		[3.04]		[−6.98]	
−ΔCDS	0.56	0.35	0.23	0.13	−0.33	0.11
	[3.71]		[2.00]		[−2.62]	
Panel B. First Principal Component of Credit Shock Variables						
Full-sample variables	0.46	0.32	0.21	0.12	−0.26	0.10
	[10.92]		[2.97]		[−5.35]	
All variables, PC average	0.44	0.33	0.20	0.12	−0.24	0.09
	[10.51]		[3.21]		[−5.73]	

9.8 Table VI: the coinsurance mechanism — covariance matters

Design: split divisions by coinsurance potential (high vs low), then regress normalized Δq on credit shocks.

Interpretation:

- Low-coinsurance divisions (high covariance / large contribution to conglomerate variance) remain sensitive to credit shocks.
- High-coinsurance divisions become nearly insensitive.
- The large "High − Low" differences support a coinsurance/internal-capital-markets channel.

9.9 Table VII: credit shocks dominate other economic shocks in explaining the wedge

Design: joint regression on an economic PC and a credit PC.

Takeaway: once credit shocks are included, the residual role of other economic shocks in explaining the divisional–standalone sensitivity wedge becomes small. This strengthens the interpretation that the "division insulation" is primarily a **financing channel**.

9.10 Table VIII: investment tracks divisional q inside conglomerates

Design: panel regression of $I = \text{CAPEX/Sales}$ on both q^{SA} and q^{Div} with interactions for conglomerate divisions.

Core implication:

- For standalone firms, investment is strongly related to standalone q and weakly (if at all) to divisional q.

- For conglomerate divisions, the opposite holds: investment is much more sensitive to divisional q than to standalone q.

Interpretation: divisional q is the relevant "opportunity set" proxy for internal allocation, consistent with the paper's conceptual goal.

Table VI

Financing Conditions: The Role of Coinsurance

This table investigates the sensitivity of qs of divisions with low and high coinsurance potential to financing conditions. It reports average slope coefficients and R^2s from industry-level univariate regressions. To estimate qs, conglomerate divisions are grouped by both their Fama-French 10-industry affiliation and their coinsurance potential, which is inversely related to a division's relative contribution to the conglomerate variance. See Section III.C in the text for more details. The dependent variable is the annual change in q, normalized to have unit standard deviation, and the independent variable is a contemporaneous credit shock. In Panel B, the independent variables are two versions of the first principal component of the variables in Panel A based on either the variables available for the entire sample or the average first principal component that uses all combinations of available variables to maximize the estimation period. t-Statistics are reported in brackets. Data sources and construction details are provided in Appendix B. The sample period is 1979 to 2018, except for ΔTED (1987 to 2018), ΔCredit rating (1986 to 2017), ΔCredit spread (2005 to 2016), and ΔCDS (1991 to 2018).

	Divisional q When Coinsurance Potential Is					
	Low		High		High – Low	
	Slope	R^2	Slope	R^2	Slope	R^2
Panel A. Credit Shock Variables						
−ΔTED spread	0.79	0.06	0.15	0.02	−0.64	0.03
	[4.59]		[1.01]		[−2.50]	
−ΔDefault spread	0.28	0.07	0.03	0.05	−0.25	0.04
	[2.39]		[0.25]		[−1.76]	
−ΔCredit rating	0.62	0.04	−0.33	0.02	−0.95	0.04
	[3.82]		[−1.90]		[−3.64]	
−ΔCredit spread	0.15	0.17	−0.09	0.10	−0.24	0.11
	[1.70]		[−0.72]		[−2.68]	
ΔDistance to default	0.21	0.17	0.03	0.03	−0.18	0.07
	[3.84]		[0.95]		[−3.27]	
−ΔCDS	0.33	0.18	−0.01	0.06	−0.34	0.09
	[2.44]		[−0.10]		[−2.91]	
Panel B. First Principal Component of Credit Shock Variables						
Full-sample variables	0.25	0.15	0.03	0.05	−0.22	0.08
	[3.50]		[0.55]		[−2.56]	
All variables, PC average	0.25	0.16	0.03	0.04	−0.22	0.08
	[3.73]		[0.53]		[−2.63]	

Table VII

Sensitivity of qs to Financing and Other Economic Conditions

This table investigates the sensitivity of standalone and divisional qs (Panel A) as well as qs of divisions with low and high coinsurance potential (Panel B) to economic and financing conditions. It reports average slope coefficients and R^2s from industry-level regressions of the annual change in q, normalized to have unit standard deviation, on contemporaneous economic and credit shocks, measured as in Panel B of Tables IV and V, respectively. In Panel A, qs are estimated for the 10 Fama-French industries. In Panel B, conglomerate divisions are grouped by both their Fama-French 10-industry affiliation and their coinsurance potential, as in Table VI. t-Statistics are reported in brackets. The sample period is 1979 to 2018.

Panel A. Standalone versus Divisional qs									
	Standalone qs			Divisional qs			Div. – SA q		
	Econ	Credit	R^2	Econ	Credit	R^2	Econ	Credit	R^2
Full-sample variables	0.06	0.40	0.34	0.04	0.16	0.13	−0.02	−0.24	0.11
	[2.12]	[6.56]		[1.72]	[1.92]		[−0.65]	[−3.87]	
All variables, PC average	0.06	0.38	0.36	0.03	0.17	0.14	−0.03	−0.21	0.11
	[2.09]	[5.95]		[1.52]	[2.18]		[−0.97]	[−3.77]	
Panel B. Divisional qs by Coinsurance Potential									
	Low			High			High – Low		
	Econ	Credit	R^2	Econ	Credit	R^2	Econ	Credit	R^2
Full-sample variables	0.03	0.21	0.17	0.01	0.02	0.06	−0.02	−0.19	0.09
	[1.51]	[2.61]		[0.59]	[0.35]		[−0.68]	[−2.46]	
All variables, PC average	0.03	0.21	0.18	0.01	0.02	0.06	−0.02	−0.19	0.10
	[1.27]	[2.75]		[0.47]	[0.37]		[−0.56]	[−2.30]	

Table VIII

The Investment-q Sensitivity of Standalone Firms and Conglomerate Divisions

This table investigates the sensitivity of investment to standalone and divisional qs. It reports results from panel regressions in which the dependent variable is the ratio of capital expenditures to sales of conglomerate divisions and stand-alone firms. The independent variables are the median q of standalone firms in the industry (Standalone q), calculated as in Berger and Ofek (1995), the divisional q in the Fama-French 10 industry (Divisional q), an indicator variable for conglomerate divisions (C), and their interactions. Columns (4) and (8) also control for the ratio of cash flows to sales. Columns (1) to (4) control for year and industry fixed effects, while columns (5) to (8) control for year and firm-division fixed effects. Standard errors are clustered by year, industry, and conglomerate status. t-Statistics are shown in brackets.

Variable	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Standalone q	0.022		0.023	0.022	0.015		0.015	0.015
	[6.31]		[6.43]	[5.94]	[5.37]		[5.30]	[5.34]
Divisional q		0.014	0.003	0.004		0.010	0.004	0.004
		[3.48]	[1.90]	[0.95]		[4.23]	[0.85]	[0.96]
C	0.010	−0.008	−0.004	−0.001	0.001	−0.008	−0.008	−0.007
	[4.93]	[−3.74]	[−1.40]	[−0.58]	[0.19]	[−1.43]	[−1.24]	[−1.14]
Standalone $q \times C$	−0.003		−0.009	−0.007	0.002		0.000	0.000
	[−1.27]		[−2.03]	[−1.49]	[0.60]		[−0.13]	[0.10]
Divisional $q \times C$		0.013	0.021	0.023		0.011	0.011	0.011
		[4.11]	[3.87]	[4.82]		[2.40]	[6.70]	[6.91]
CFS				−0.041				−0.058
				[−2.67]				[−3.61]
$CFS \times C$				−0.062				−0.010
				[−3.50]				[−0.68]
Year fixed effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Industry fixed effects	Yes	Yes	Yes	Yes				
Firm-division fixed effects					Yes	Yes	Yes	Yes
R^2	0.391	0.386	0.391	0.397	0.72	0.718	0.72	0.722
Number of observations	116,813	116,813	116,813	116,813	110,120	110,120	110,120	110,120

9.11 Table IX: spinoffs — investment sensitivities switch with organizational form

Design: around spinoffs, compare pre vs post using the PRE interactions.

Interpretation:

- Pre-spinoff (division-like), investment loads positively on divisional q and less on standalone q.
- Post-spinoff (standalone-like), investment becomes more strongly related to standalone q.

This is a clean within-entity piece of evidence that the paper's q^{Div} captures information tied to organizational structure.

9.12 Table X: RDQ predicts diversifying acquisition activity

Design: regress log(1+# deals) on lagged standalone q and lagged RDQ.

Interpretation: when the "relative value of conglomeration" is high (higher RDQ), the economy reallocates assets across firm boundaries via more diversifying M&A.

9.13 Table XI: RDQ predicts announcement returns of diversifying acquisitions

Design: regress deal CARs on lagged q and lagged RDQ (plus deal-party controls).

Takeaway: RDQ positively predicts announcement returns for acquirer/target/combined, consistent with the market viewing diversification as more value-creating when divisional valuations exceed standalone valuations.

10 Big-picture interpretation

I connect the results as follows:

Table IX
The Investment-q Sensitivity around Spinoffs

This table investigates the sensitivity of investment to standalone and divisional qs around the spinoffs of conglomerate divisions. It reports results from regressions in which the dependent variable is the ratio of capital expenditures to sales of the affected entities. The independent variables include the median q of standalone firms in the industry (Standalone q), calculated as in Berger and Ofek (1995), the divisional q in the Fama-French 10 industry (Divisional q), an indicator variable for the prespinoff period (*PRE*), and their interactions. For each spinoff, we include observations from three years before the spinoff to three years after the spinoff. Column (4) restricts the sample to spinoffs with at least one nonmissing investment observation in both the pre- and postspinoff periods, and column (5) requires at least two observations in each period. All regressions include year and industry fixed effects. Standard errors are clustered by industry and year. t-Statistics are shown in brackets.

	Firm Obs Required Both Pre- and Postspinoff				
	No Restriction			≥ 1	≥ 2
	(1)	(2)	(3)	(4)	(5)
Standalone q	0.040		0.044	0.049	0.052
	[3.39]		[3.71]	[2.41]	[2.26]
Divisional q		0.014	−0.009	0.005	0.017
		[0.79]	[−0.63]	[0.23]	[0.35]
PRE	0.023	−0.036	−0.016	−0.025	−0.047
	[1.26]	[−2.51]	[−1.42]	[−1.15]	[−2.50]
Standalone q × *PRE*	−0.012		−0.026	−0.027	−0.028
	[−1.47]		[−3.13]	[−1.80]	[−1.70]
Divisional q × *PRE*		0.032	0.051	0.062	0.080
		[2.51]	[2.84]	[2.09]	[2.90]
R^2	0.137	0.113	0.141	0.148	0.203
Number of observations	1,551	1,551	1,551	1,057	711

Table X
Acquisition Activity

This table reports results of panel regressions of the logarithm of one plus the number of diversifying acquisitions in an industry-year on the median q of standalone firms and the relative divisional q (RDQ) in the Fama-French 10 industry. The sample requires both target and acquirer to be public standalone firms, and the acquisition to be diversifying across two three-digit SIC industries. All regressions include industry fixed effects. Standard errors are clustered by industry. t-Statistics are shown in brackets.

	Standalone q	RDQ	R^2
Number of acquirers by industry	0.329	0.437	0.572
	[3.65]	[3.39]	
Number of acquisition targets by industry	0.308	0.327	0.629
	[2.81]	[2.27]	
Number of targets or acquirers by industry	0.316	0.380	0.570
	[2.64]	[2.76]	

Table XI
Acquisition Announcement Returns

This table analyzes the relation between qs and acquisition announcement returns. Each column reports estimates from a regression in which the dependent variable is the acquisition announcement return of acquirers, targets, or the combined firm (i.e., the value-weighted average). Returns are cumulated over the five-day period centered around the announcement date and are in excess of market returns. The independent variables include the median q of standalone firms in the industry, calculated as in Berger and Ofek (1995), as well as the relative divisional q (RDQ). All regressions also control for log market capitalization, profitability, and asset growth of both target and acquirer. All independent variables are measured in the year prior to the announcement year. The sample requires both target and acquirer to be public standalone firms, and the acquisition to be diversifying across 2 three-digit SIC industries (columns (1), (2), (4), (5), and (7)) or across two Fama-French 10 industries ((3), (6), (8)). All regressions include year fixed effects and cluster standard errors by year. t-Statistics are shown in brackets.

	Dependent Variable							
	Acquirer Return			Target Return			Combined Return	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Standalone q of the acquirer	0.021							
	[3.93]							
Standalone q of the target				−0.006				
				[−0.26]				
Standalone q of the combined firm		0.019	0.025		0.043	0.070	0.016	0.027
		[2.71]	[2.68]		[2.70]	[3.14]	[2.74]	[3.85]
RDQ of the acquirer	0.056							
	[3.69]							
RDQ of the target				−0.056				
				[−1.42]				
RDQ of the combined firm		0.050	0.055		0.064	0.100	0.045	0.056
		[2.62]	[3.46]		[1.61]	[1.77]	[2.39]	[4.22]
SIC three-digit diversification	Yes	Yes		Yes	Yes		Yes	
Fama-French 10 diversification			Yes			Yes		Yes
R^2	0.204	0.174	0.297	0.199	0.201	0.307	0.175	0.322
Number of observations	336	336	143	336	336	143	336	143

1. The new estimator produces divisional q's that are *empirically distinct* from standalone q's (Table II, Figure 2) and less noisy in conglomerate valuation decompositions (Figure 1).

2. Divisional q's are less sensitive to shocks, especially credit shocks (Tables IV–V), implying divisions are insulated from external financing conditions.

3. This insulation is strongest where coinsurance is feasible (low covariance contributions) (Table VI) and the wedge is largely explained by credit conditions (Table VII).

4. These wedges are *real*: they show up in internal investment allocation (Tables VIII–IX) and in the frequency and returns of diversifying acquisitions (Tables X–XI).

In one line. The paper turns divisional valuation measurement from a "standalone proxy" problem into a "conglomerate-implied" estimation problem and uses it to map financing frictions and covariance structures into both internal and external asset allocation.

11 Conclusion

11.1 Summary

This paper concludes that **the standard practice of valuing conglomerate divisions using standalone-firm industry multiples is systematically misleading**, and that **a conglomerate-implied approach reveals economically important wedges in valuations that map into financing frictions, internal capital markets, and real asset reallocation**.

The paper's logic is coherent and proceeds in one chain:

1. **Measurement innovation: estimate division-level valuation multiples without standalone firms.** The authors propose a new method to estimate *divisional* valuation multiples ("divisional q's") using only conglomerates and their segment fundamentals. Intuitively, the cross-section of conglomerates provides enough variation in segment shares to identify class-level multiples via year-by-year median (quantile) regressions. This avoids a strong (and often false) assumption that standalone-firm q is a valid proxy for the q of the corresponding conglomerate division.

2. **Divisional q's differ materially from standalone q's across industries and over time.** Divisional valuations range from *deep discounts* to *large premia* relative to standalone benchmarks, and the relationship between divisional and standalone q is highly industry-dependent and state-dependent. Over time, divisional q's are **about 17% less volatile** than standalone q's, suggesting meaningful differences in how shocks transmit into valuations.

3. **Mechanism: internal capital markets and coinsurance mitigate external financing frictions.** A central empirical conclusion is that divisional q's are **less sensitive to economic and, especially, credit shocks** than standalone q's. Using multiple credit-shock measures (e.g., TED spread, ratings-based measures, CDS spreads), the sensitivity of divisional q's to credit shocks is, on average, **about 54% lower** than for standalone firms. This reduced sensitivity is strongest for divisions that have *low covariance* with the rest of the conglomerate (high coinsurance potential), for which credit-shock sensitivity becomes statistically indistinguishable from zero. Credit shocks also absorb much of the role of broader economic shocks in explaining the divisional–standalone wedge, implying a financing channel.

4. **Real implications: divisional q's map into internal and external resource allocation.** The paper shows that divisional q's are the relevant "opportunity" signals inside conglomerates: conglomerate investment and internal capital allocation are much more sensitive to divisional q than to standalone q, and this sensitivity flips around spinoffs in a manner consistent with organizational form changing the relevant valuation benchmark. Moreover, differences between divisional and standalone q (the RDQ wedge) predict **cross-firm reallocation** through diversifying M&A: industries with higher RDQ experience more diversifying acquisitions, and RDQ positively predicts announcement returns of diversifying deals. Quantitatively, the paper reports that a one-standard-deviation increase in the combined-firm RDQ is associated with about a **0.8%** higher combined-firm announcement return.

11.2 Why divisional q differs from standalone q

The intuition can be summarized as **organizational form changes the effective pricing of fundamentals because it changes exposure to financing constraints and the ability to reallocate resources**.

1) Why standalone multiples are not a clean proxy. Standalone firms and conglomerate divisions differ along dimensions that are hard to observe but crucial for valuation: access to internal funding, risk-sharing across divisions, and the ability to smooth shocks via internal capital markets. Even within the same industry classification, a division inside a conglomerate is not "the same asset" as a standalone firm.

2) Coinsurance lowers a division's effective shadow cost of external finance. When external credit tightens, standalone firms that rely on external markets face a higher marginal cost of capital, lowering valuations. Conglomerates can shift resources internally, so divisions are partially insured against external financing shocks. This implies:

$$\text{tighter credit} \;\Rightarrow\; \text{standalone valuation drops more} \;\Rightarrow\; \text{divisional–standalone wedge widens.}$$

3) Covariance structure determines how much insurance is feasible. Internal coinsurance works best when a division's cash flows (or fundamentals) are weakly correlated with the rest of the conglomerate. Low covariance means that when the rest of the conglomerate is doing well, it can support the division; when the division is doing well, it can support others. Hence, covariance structure is a sufficient statistic for the *strength* of the internal-capital-market buffer, which is exactly what the paper confirms in the credit-shock tests.

4) Why RDQ predicts investment and diversifying acquisitions. RDQ measures the *relative* valuation of assets inside conglomerates versus as standalone entities. When RDQ is high, the organizational form "conglomerate" makes those assets more valuable—especially because internal capital markets are more valuable in that state. Then it is efficient (and value-creating) to reallocate assets toward conglomerate ownership:

$$\text{RDQ} \uparrow \Rightarrow \text{ greater incentive for diversification } \Rightarrow \text{ more diversifying M\&A and higher announcement gains.}$$

11.3 Takeaway

Boguth, Duchin, Simutin (2022) show that division-level valuation multiples implied by conglomerates differ substantially from standalone benchmarks and are less volatile and much less credit-shock-sensitive (about 54% lower sensitivity), consistent with covariance-driven coinsurance and internal capital markets; these wedges matter for real allocation inside firms (investment) and across firms (diversifying M&A), so their conglomerate-implied divisional q provides a more economically meaningful measurement tool than standalone-based proxies.